

EX. NO: 6 IMPLEMENTATION OF DIFFIE–HELLMAN KEY EXCHANGE ALGORITHM

AIM

To implement the Diffie–Hellman Key Exchange Algorithm using Python.

DESCRIPTION

The Diffie–Hellman Key Exchange Algorithm is a public-key cryptographic protocol that enables two users to securely establish a shared secret key over an insecure communication channel. The shared secret key can then be used for secure communication using symmetric encryption algorithms.

ALGORITHM

STEP-1: Select two public numbers, a prime number **P** and a primitive root **G**.

STEP-2: User A selects a private key **a**.

STEP-3: User B selects a private key **b**.

STEP-4: User A computes **A = G^a mod P**.

STEP-5: User B computes **B = G^b mod P**.

STEP-6: Exchange the public values **A** and **B**.

STEP-7: User A computes the shared key as **K = B^a mod P**.

STEP-8: User B computes the shared key as **K = A^b mod P**.

STEP-9: Verify that both shared keys are equal.

CODE

```
# Diffie-Hellman Key Exchange Algorithm
```

```
# Public values
```

```
P = int(input("Enter Prime Number (P): "))
```

```
G = int(input("Enter Primitive Root (G): "))
```

```
# Private keys
```

```
a = int(input("Enter Private Key of User A: "))
```

```
b = int(input("Enter Private Key of User B: "))
```

```
# Public keys

A = pow(G, a, P)
B = pow(G, b, P)

print("\nPublic Key of User A:", A)
print("Public Key of User B:", B)

# Shared Secret Keys

keyA = pow(B, a, P)
keyB = pow(A, b, P)

print("\nSecret Key for User A:", keyA)
print("Secret Key for User B:", keyB)

if keyA == keyB:
    print("\nKey Exchange Successful!")
else:
    print("\nKey Exchange Failed!")
```

Output

[Clear](#)

```
Enter Prime Number (P): 23
Enter Primitive Root (G): 5
Enter Private Key of User A: 6
Enter Private Key of User B: 15

Public Key of User A: 8
Public Key of User B: 19

Secret Key for User A: 2
Secret Key for User B: 2

Key Exchange Successful!

=== Code Execution Successful ===
```

RESULT

Thus, the Diffie–Hellman Key Exchange Algorithm was implemented successfully using Python.